Fiscal Year:	FY 2017	**Task Last Updated:**	FY 04/15/2017
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Project Title:	High-Precision Microwave Spectroscopy of Long-Lived Circular-State Rydberg Atoms in Microgravity		
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Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
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Comments:			
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Key Personnel Changes/Previous PI:	Jan 2017 report: No changes.		
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Objective: The system of fundamental constants (the speed of light, Planck's constant, the fine structure constant, the Rydberg constant, etc.) is important because it directly reflects our current understanding of the foundations of nature. The objective of the present project is to measure one of these constants, the Rydberg constant, at an uncertainty level in the range of a few parts in 10^12 (the current uncertainty stands at 5 parts in 10^12). It is planned to use circular Rydberg atoms, which are very large, highly excited atoms with doughnut-shaped electron distributions that do not penetrate into the atomic nucleus. As a result, these atoms are not susceptible to perturbations caused by electron wavefunction overlap with the nuclear charge density and quantum-electrodynamics (QED) corrections. It is this property that makes a measurement of the Rydberg constant with circular Rydberg atoms particularly useful and potentially highly competitive, in comparison with already established methods. Due to the absence of critical QED corrections and nuclear penetration, a measurement of the Rydberg constant with circular Rydberg atoms would help

Task Description:	resolve a recently reported inconsistency, where it was concluded that "... either the Rydberg constant has to be shifted by -110kHz/c (4.9 standard deviations), or the calculations of the QED [quantum-electro-dynamic] effects in atomic hydrogen or muonic hydrogen atoms are insufficient." [R. Pohl et al., Nature 466, 213 (2010)]. Method: Rb atoms are cooled to below 1 microK and laser-excited into a low-angular-momentum Rydberg level with principal quantum number n on the order of 40. Using an adiabatic atomic-state transformation method, the Rydberg atoms are then transferred into the circular state |n,n1=0,n2=0,m=n-1> (m, n1, and n2 are the magnetic and parabolic quantum numbers). An optical lattice (wavelength 1064 nm or 532 nm on the ground, 850 nm at CAL) is adiabatically ramped on. The Rydberg atoms are trapped in the lattice via the ponderomotive interaction of the Rydberg electron with the lattice light. During the hold time of the atom cloud in the lattice, the electric-quadrupole transition into the state |n+2,n1=1,n2=1,m=n-1> is driven by amplitude-modulating the optical lattice at the transition frequency. The transition is probed using state-selective electric-field ionization. The measured transition frequency allows one to extract the Rydberg constant. Key advantages of the method are that the transition is free of first-order Stark and Zeeman shifts, that QED, nuclear-penetration, hyperfine and other shifts are small, and that the optical lattice causes only moderate trap-induced transition shifts. Theoretical work includes the modeling of the circular-state preparation using multi-level adiabatic passage and of the circular-state transition dynamics in a shallow, microwave-modulated optical lattice, in which center-of-mass wave-packet expansion and tunneling must be considered. Relevance to NASA and relevance of microgravity: High-precision measurement of fundamental constants represents an important part of contemporary atomic-physics research. The proposed high-precision measurement of the Rydberg constant would add a new component to the Cold Atom Laboratory (CAL) research portfolio. To achieve optimal spectroscopic performance it is critical to reduce systematic lattice-induced transition shifts and related shifts. Since the most critical shifts diminish with decreasing optical-lattice depth, the lattices must be chosen as shallow as possible. Microgravity conditions are important because they enable the use of shallow lattices without the atoms falling out. The microgravity environment provided by the CAL, including its 850 nm optical-lattice laser, offers an ideal platform for the project to reach its ultimate uncertainty goal. The CAL setup may also provide an extended period of measurement time to acquire good statistics in the high-precision spectra.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	High-precision spectroscopy and metrology impacts fields such as time measurement and navigation. Benefits to the general public include atomic clocks and the GPS system. The project “High-Precision Microwave Spectroscopy of Long-Lived Circular-State Rydberg Atoms in Microgravity” involves the development of a fundamentally new type of spectroscopy, which has been successfully demonstrated during the first year of the grant. The future development of laser systems to hone this innovative technique may lead to benefits in other disciplines, such as chemistry, biology, and engineering. The research involves graduate students, undergraduate students, and post-docs, who are being prepared for future jobs in academia, government, and industry. Having a capable scientific workforce is essential for progress and competitiveness in science and technology, as well as for the national economy. The main goal of the project, which is to measure the Rydberg constant with circular Rydberg atoms, will help resolve the “proton radius puzzle.” The puzzle has arisen from inconsistencies in measurements on low-lying electronic and muonic states of hydrogen, which involve the electromagnetic interaction, QED, the proton structure, and QCD. The project will therefore serve as an indirect test for QED and the structure of the proton. With Rydberg atoms, the Rydberg constant will be tested at electron energies in the range of milli-electronvolts. This regime is rather different from the energy regime of previous measurements (eVs to keVs).
	Knowing the value of the Rydberg constant accurately has been of interest for decades due to its relation to other fundamental constants and its role in calculations of atomic energy levels. More recently, a large discrepancy in the proton radius has been found using muonic hydrogen, known as the proton radius puzzle [https://], leading to various more or less wild speculations. The Quanta Magazine reported on August 11, 2016: “This [new physics] “would, of course, be fantastic,” said Randolf Pohl of the Max Planck Institute of Quantum Optics in Garching, Germany, who led both the 2010 experiment and the new study. “But the most realistic thing is that it’s not new physics.” ”. And it is further reported “His [Pohl’s] personal guess is that physicists have misgauged the Rydberg constant, a factor that goes into calculating the expected differences between atomic energy levels.” SOURCE: https:// So, as of August 2016, the proton radius puzzle has gotten worse. This situation has reinforced the need to confirm the accuracy of the Rydberg constant. Previous precision experiments with this goal have involved low-lying states of hydrogen, limited typically by statistical uncertainties and second-order Doppler shifts. These have led to the best current available relative uncertainty for the Rydberg constant value of 5.9 x 10^-12 [Codata]. There has also been a study involving circular Rydberg states of hydrogen (relative uncertainty of 2.1 x 10^-11, by deVries and Kleppner et al.) and a proposal involving circular states of lithium, with an expected relative uncertainty of about 10^-10 (Haroche and coworkers). The main limitations for the former have been the presence of a non-zero Zeeman shift (2.6 Hz), short interaction times and a non-negligible second-order Doppler shift (1.4 Hz). The approaches involving low-lying states and circular states deal with significantly different frequency regimes: optical versus microwave. Therefore, measurements involving low-lying states of hydrogen can, in principle, have a better relative uncertainty than results for circular states (under the assumption of similar absolute uncertainty). However, circular states are insensitive to critical systematics that are limiting in spectroscopy of low-lying states. These are the nuclear penetration (the origin of the proton radius puzzle) and QED shifts. The project “High-Precision Microwave Spectroscopy of Long-Lived Circular-State Rydberg Atoms in Microgravity” deals with measuring the Rydberg constant and other atomic constants, such as polarizabilities, using long-lived circular Rydberg atoms. The large size of the Rydberg atoms allows one to employ a fundamentally new method of spectroscopy. The method is based on harnessing a term known as the “A-square term” for the purpose of spectroscopy. The method requires one to have a modulated light field with a spatial modulation period that is on the same order as the diameter of the atom that is being probed. At the same time, the field must be periodically modulated in time, with a

Task Progress:	modulation frequency that matches the frequency of the atomic transition that is to be probed. Rydberg atoms in electro-optically modulated standing wave light fields satisfy both conditions. A manuscript, with an exhaustive list of expected systematics, is close to submission. This manuscript will present the project to the wider scientific community. Some useful new input may result from this. In extracting the Rydberg constant, the polarization quantum defect and the core polarizability of rubidium, the atomic species used in the work, must be known better than it is currently known (by up to a factor of ten, depending which published numbers one uses). The core polarizability accounts for the fact that the atomic core (the Rb+ ion in this case) becomes electrically polarized by the Rydberg electron that orbits the Rb+ ion. The core polarization leads to Rydberg level shifts on the order of 10 kHz. The small, simplified spectroscopy chamber set up in 2015 in the current project has been used in 2016 to perform a precision measurement of the rubidium g-level quantum defects. This work has been completed and is close to submission. One graduate student has, in large parts, graduated with this result. In her work, results are compared with another recent measurement, which is by Gallagher et al. [J. Lee, J. Nunkaew, and T. Gallagher, Physical Review A 94, 022505 (2016)]. The measurements are consistent with each other, and the present one (ours) is about one order of magnitude more precise. In the last year, the team has continued to work on a new cryogenic spectroscopic setup that is supposed to last through the entire performance period of about 3 more years. The setup features a three-dimensional optical lattice for Rydberg atom trapping and lattice modulation spectroscopy. It incorporates all ingredients necessary for preparing the above-mentioned circular Rydberg states, which will be required to reach the desired spectroscopic accuracy and precision. The circular-state production method follows pioneering work by D. Kleppner et al. at MIT [R. G. Hulet and D. Kleppner, Physical Review Letters 51, 1430 (1983)] and Haroche at ENS [P. Nussenzveig, F. Bernardot, M. Brune, J. Hare, J. M. Raimond, S. Haroche, and W. Gawlik, Phys. Rev. A 48, 3991 (1993)]. The setup is ready for cryogenic operation at 4 Kelvin, which will be required in order to suppress unwanted transitions and lifetime reductions due to black-body thermal radiation. The setup also includes the required laser-cooling infrastructure as well as control of electric and magnetic fields within the spectroscopic volume. The vacuum setup and its complicated internal electrode structure has been completed, baked out, and integrated with lasers. The setup has been fitted with a primary-MOT glass cell from Precision Glass Blowing. The primary MOT, a 2D+ beam MOT, and the secondary MOT inside the spectroscopic enclosure have been set up and are operated on a near-daily basis. The secondary MOT can also be run in molasses mode, which will be required in upcoming experiments with very well controlled magnetic fields. The experimental work is accompanied by a theoretical effort conducted by V. Malinovsky. The present objective of the theory work is to model the spectroscopic line shapes of Rydberg transitions in an amplitude-modulated Rydberg-atom lattice in a manner that takes the quantization of the center-of-mass motion into account. This ability will be an important ingredient to determine the line center with 1Hz uncertainty. To take into account the quantization of motion of the Rydberg atoms in the optical lattice, a theoretical model is developed in which both ground and excited wave functions are subject to the periodic potentials, representing the lattice fields, while the coupling (the effective Rabi frequency) is also periodic as function of the translational coordinate. To obtain the spectrum of the excited-state population, the time-dependent Schroedinger equation has been solved in the momentum representation. Two manuscripts on the theory work are in preparation. References J. De Vries. A precision millimeter-wave measurement of the Rydberg frequency, PhD thesis, (2001). J. Hare, A. Nussenzweig, C. Gabbanini, M. Weidemueller, P. Goy, M. Gross, and S. Haroche. IEEE Transactions on Instrumentation and Measurement, 42, 331 (1993).
Bibliography Type:	Description: (Last Updated: 02/24/2025)
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